

1. Dataset Collection & Corpus Definition

1.1 Purpose

The purpose of dataset collection is to establish a **reliable, domain-specific, traceable, and controlled knowledge corpus** for the Insurance RAG AI Assistant.

The system is designed to answer insurance-related questions using evidence retrieved from the selected policy documents rather than relying exclusively on the pretrained knowledge of the underlying Large Language Model (LLM).

The dataset therefore forms the **authoritative knowledge boundary** of the initial system.

The collection process is designed to ensure that:

- the selected documents are relevant to the intended insurance use case;
 - each document can be associated with a specific insurer and product;
 - the source of policy information can be traced;
 - different products are represented independently;
 - unnecessary duplication is avoided;
 - the corpus remains extensible for future products and policy versions.
-

1.2 Business Problem Addressed by the Dataset

Insurance products contain information distributed across lengthy and highly detailed policy documents.

Relevant information may include:

- coverage and benefits;
- eligibility;
- definitions;
- waiting periods;
- exclusions;
- financial limits and sub-limits;
- optional covers;
- deductibles and co-payments;
- claim conditions;
- renewal conditions;
- portability and migration rules;
- product-specific clauses and exceptions.

A general-purpose LLM may understand the general concept of health insurance but cannot be assumed to know the exact contractual conditions of a particular insurance product.

For example, the Star Comprehensive documentation specifies a **36-month waiting period for pre-existing diseases**, while separately providing an optional mechanism to reduce the waiting period to **12 months**.

Therefore, the system requires access to the **actual product-specific source documents**.

1.3 Dataset Objective

The dataset is designed to support four primary capabilities:

1. Product-specific Question Answering

Example:

"What is the waiting period for pre-existing diseases under Star Comprehensive?"

2. Evidence-based Question Answering

Example:

"What expenses are covered before hospitalization?"

3. Cross-product Comparison

Example:

"Compare the pre-existing disease waiting periods across the available policies."

4. Product-aware Retrieval

The system must distinguish between information belonging to different insurers and products.

For example:

```
User Query
↓
Identify relevant product(s)
↓
Retrieve evidence
↓
Verify product/source
↓
Generate answer
```

1.4 Corpus Definition

The initial corpus consists of **five insurance products from five different insurers**.

ID	Insurance Provider	Product	Corpus Role
INS-001	HDFC ERGO	my:Optima Secure	Primary policy source
INS-002	Niva Bupa	ReAssure 2.0	Primary policy source
INS-003	Care Health Insurance	Ultimate Care	Primary policy source
INS-004	ICICI Lombard	Elevate	Primary policy source
INS-005	Star Health and Allied Insurance	Star Comprehensive	Primary policy source

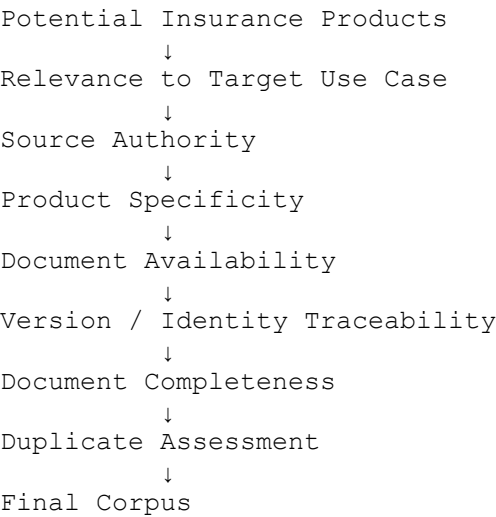
The five-product corpus is intentionally limited for the initial implementation and evaluation.

It is **not intended to statistically represent the entire insurance industry**.

1.5 Corpus Selection Strategy

The corpus was constructed using a **controlled selection strategy** rather than collecting documents randomly.

The selection process is:



This ensures that every selected document has a defined role in the system.

1.6 Selection Criteria

A document is considered suitable for inclusion when it satisfies the following criteria.

1.6.1 Domain Relevance

The document must belong to the insurance domain targeted by the project.

1.6.2 Product Relevance

The document must correspond to one of the explicitly selected insurance products.

1.6.3 Source Authority

Preference is given to documents originating from the respective insurer or an authoritative public source.

1.6.4 Product Specificity

The document must contain product-specific information rather than only generic insurance education.

1.6.5 Information Depth

The document should contain sufficient detail to support meaningful question answering and comparison.

1.6.6 Traceability

The source must allow retrieved information to be associated with the originating document and product.

1.6.7 Version Identification

Where available, identifiers such as UIN, version, policy number reference, or document date are retained.

For example, the Star Comprehensive documents identify the product as **Star Comprehensive Insurance Policy** with UIN **SHAHLP26044V092526**.

1.7 Why Real Insurance Documents Were Selected

The primary corpus uses **real publicly available insurance policy documents**.

This is a deliberate engineering decision.

The project is intended to evaluate the ability of a RAG system to operate on **naturally occurring enterprise-style documents**, rather than documents whose structure was artificially designed for the experiment.

Real policy documents contain characteristics such as:

- different terminology;
- different section structures;
- different numbering conventions;
- conditional clauses;
- exceptions;
- exclusions;
- waiting periods;
- financial limits;
- tables;
- cross-references;
- product-specific benefits.

For example, the Star Comprehensive documentation contains detailed coverage information for hospitalization, day-care treatment, AYUSH, modern treatments, pre-hospitalization, post-hospitalization, home care, cumulative bonus, automatic restoration, delivery/newborn coverage and other benefits.

This provides a realistic knowledge environment for evaluating the proposed system.

1.8 Why Synthetic Data Was Not Selected as the Primary Corpus

Synthetic policy documents were considered unsuitable as the primary dataset.

Synthetic documents can be useful for controlled unit testing, but they have a major limitation:

Their structure and content are known and controlled by the developer.

For example, a synthetic policy could intentionally be created as:

```
Policy
├── Coverage
├── Waiting Period
├── Exclusions
├── Claims
└── Renewal
```

This makes document processing and retrieval significantly easier.

The proposed project, however, needs to demonstrate that the system can handle **real document variability**.

Therefore:

Synthetic Corpus	Real Policy Corpus
Developer-controlled structure	Naturally occurring structure
Predictable terminology	Product-specific terminology
Easy document consistency	Document variability
Easy chunk boundaries	Complex document boundaries
Limited real-world exceptions	Real policy conditions and exceptions
Useful for controlled testing	Suitable for realistic system evaluation

Final decision

Real policy documents are used as the primary corpus.

Synthetic documents may be introduced later for:

- controlled unit tests;
- edge-case testing;
- failure-case generation;
- regression testing.

They are not used as the primary knowledge source.

1.9 Why Five Products Were Selected

The project does not aim to build a single-document question-answering system.

It is intended to support **multi-product insurance information retrieval and comparison**.

A single policy would allow us to test:

"What does this policy say?"

But it would not adequately test:

"How does this policy differ from another policy?"

Therefore, multiple products are required.

Five products provide an initial balance between:

- corpus diversity;

- manageable development scope;
- meaningful cross-product comparison;
- retrieval evaluation;
- implementation complexity.

The corpus can later be expanded without redesigning the overall dataset strategy.

1.10 Why Products From Different Insurers Were Selected

The selected products come from **different insurance providers**.

This is intentional.

The project assumes that different insurers may define similar insurance concepts differently.

For example, two policies may both discuss:

"Pre-existing disease"

but may differ in:

- waiting period;
- conditions;
- exceptions;
- applicability;
- optional modifications.

Therefore, the system must not treat insurance terminology as universally interchangeable.

Instead, the source product must remain identifiable throughout the system.

The dataset therefore introduces a deliberate **cross-insurer information variation**.

1.11 Why We Selected One Primary Policy Document Per Product

Insurance providers commonly publish several documents for a product:

- Policy wording;
- Prospectus;
- Customer Information Sheet;

- Brochure;
- Claim forms;
- Product guides.

For the initial corpus, the project uses the **detailed policy wording / policy terms and conditions as the primary source.**

Reason

The detailed policy document contains the product-specific conditions needed for reliable policy-level question answering.

The Star Comprehensive Customer Information Sheet explicitly identifies itself as a key-information document and advises the policyholder to refer to the policy document.

Therefore, the initial corpus follows:

One product → one primary authoritative policy document.

Supporting documents are not included merely because they are available.

They will only be added when a specific system requirement justifies their inclusion.

1.12 Why We Do Not Collect Every Available Document

Collecting every document published by an insurer can introduce:

- duplicate information;
- conflicting summaries;
- multiple representations of the same clause;
- unnecessary retrieval candidates;
- increased corpus maintenance requirements.

For example, the same benefit may appear in both a prospectus and a customer information sheet.

Rather than increasing corpus size without a defined purpose, the initial implementation establishes a **controlled knowledge boundary.**

This improves:

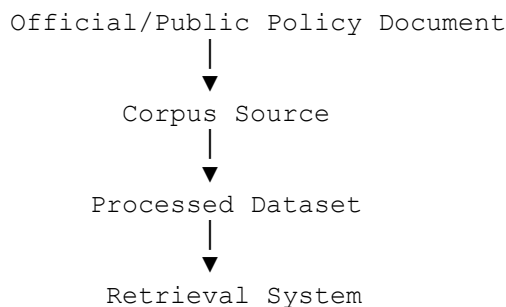
- source traceability;
- retrieval interpretation;
- corpus maintenance;
- evaluation consistency.

1.13 Source-of-Truth Strategy

The original policy documents are retained as the **source-of-truth layer**.

They are not manually rewritten or replaced with generated summaries.

The conceptual relationship is:



The processed dataset must remain traceable to the original document.

This is particularly important for conditional insurance clauses.

For example, the Star Comprehensive policy contains both a general 36-month PED waiting period and an optional reduction to 12 months.

A manually simplified dataset could accidentally retain only one of these facts and produce a misleading answer.

Therefore, **the original source is always retained for verification**.

1.14 Document Identity and Metadata

Each corpus document is associated with explicit identity metadata.

The dataset will maintain information such as:

Metadata	Purpose
document_id	Unique internal identifier
insurer	Identifies insurance provider
product_name	Identifies insurance product
UIN	Product identification where available
document_type	Policy wording / terms and conditions
version	Document version where available

Metadata	Purpose
source	Origin of document
collection_date	Dataset collection tracking
status	Active corpus document / superseded document

This prevents the system from treating similarly named documents as interchangeable.

1.15 Data Governance

Because the project is based on an industry-inspired insurance use case, data governance is explicitly defined.

Included

- Publicly accessible insurance documents
- Product-level policy information
- Public policy terms and conditions
- Public product documentation

Excluded

- Customer personal information
- Policyholder records
- Claim records
- Internal company documents
- Confidential business information
- Proprietary internal datasets
- Customer-specific policy information

The project therefore demonstrates an enterprise insurance AI workflow **without exposing confidential information**.

1.16 Dataset Assumptions

The following assumptions define the initial corpus.

A1 — Public Source Assumption

The collected documents are publicly accessible insurance documents.

A2 — Product Identity Assumption

Each selected document corresponds to the intended product.

A3 — Authority Assumption

The selected primary policy document is treated as the authoritative source for the corresponding product/version.

A4 — Version Assumption

The collected document represents the version available at the time of dataset collection.

A5 — Product Variation Assumption

Different insurers may define similar insurance concepts differently.

A6 — Limited Corpus Assumption

Five products are sufficient for initial development and evaluation but do not represent the complete insurance market.

A7 — Traceability Assumption

Information used by the system must ultimately be traceable to its originating document.

1.17 Dataset Risks

Risk	Impact	Mitigation
Policy version changes	Answers may become outdated	Preserve version/UIN and collection date
Multiple documents for same product	Duplicate evidence	Select a primary source
Different insurer terminology	Retrieval difficulty	Multi-product corpus
Complex policy conditions	Incorrect simplification	Preserve original source
Limited number of products	Limited generalization	Expand corpus in later versions
Public document availability changes	Reproducibility issues	Preserve collected source artifacts
Similar product names	Source confusion	Explicit product metadata

1.18 Dataset Limitations

The dataset has several known limitations.

Limited Product Coverage

Only five insurance products are included initially.

Limited Market Representation

The selected products do not represent every insurer, product category, or policy available in the market.

Version Dependence

Insurance policies can change over time. Therefore, the dataset represents the versions collected during the project.

Public-Source Dependence

The project relies on publicly available documents and does not incorporate confidential insurer-side information.

No Customer-Specific Data

The initial corpus represents **product-level policy information**, not individualized insurance contracts.

These limitations are explicitly acknowledged rather than hidden.

1.19 Corpus Versioning

The dataset will be version-controlled.

Example:

```
Corpus Version: v1.0
Corpus Type: Insurance Policy Corpus
Products: 5
Primary Documents: 5
Source Type: Public Policy Documents
Status: Development / Evaluation
```

If a policy document is replaced in the future:

```
v1.0
↓
Policy version updated
↓
v1.1
```

The change should be recorded rather than silently replacing the previous source.

This allows future evaluation results to identify **which corpus version produced a particular result**.

1.20 Dataset Acceptance Criteria

A document is admitted into the corpus only when the following conditions are satisfied:

- ☒ Relevant to the target insurance use case
 - ☒ Belongs to the selected product
 - ☒ Source is publicly accessible and identifiable
 - ☒ Product identity can be established
 - ☒ Policy-specific information is present
 - ☒ Document can be retained as an original source
 - ☒ Version/UIN information is recorded where available
 - ☒ Duplicate documents are evaluated
 - ☒ Document has a defined role in the corpus
-

1.21 Dataset Collection Output

At the completion of this phase, the project will maintain:

```
dataset/  
├── raw/  
│   ├── HDFC_ERGO/  
│   │   └── my_optima_secure.pdf  
│   ├── NIVA_BUPA/  
│   │   └── reassure_2.pdf  
│   ├── CARE_HEALTH/  
│   │   └── ultimate_care.pdf  
│   ├── ICICI_LOMBARD/  
│   │   └── elevate.pdf  
│   └── STAR_HEALTH/  
│       └── star_comprehensive.pdf
```

At this stage, these files represent the **original source corpus**.

No chunking, embedding, vector database, retrieval, or LLM processing is part of this phase.

1.22 Dataset Collection Completion Definition

Dataset Collection is considered complete when:

1. The target products have been explicitly defined.
 2. The five selected products have been collected.
 3. Each product has an identified primary policy document.
 4. Original source documents have been preserved.
 5. Product and insurer identity has been recorded.
 6. Available UIN/version information has been recorded.
 7. Source and collection information has been recorded.
 8. Duplicate/supporting documents have been evaluated.
 9. Dataset assumptions and limitations have been documented.
 10. Corpus governance and versioning rules have been established.
-

1.23 Final Corpus Design Decision

The final decision for Phase 1 is:

The Insurance RAG system will initially use a controlled corpus of five publicly available, product-specific insurance policy documents sourced across five different insurers. One primary policy document is maintained for each selected product, while the original documents are preserved as the source of truth. Synthetic data is excluded from the primary knowledge corpus because it does not adequately represent real-world policy variability. The corpus is deliberately limited in size for controlled development and evaluation, while its design allows future expansion to additional insurers, products, and policy versions.